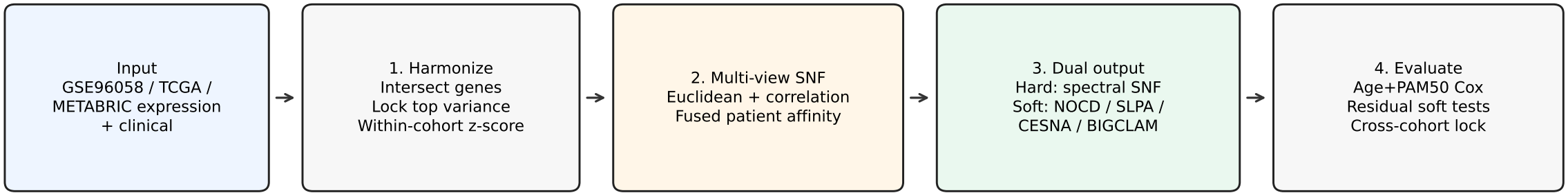Dual-output patient-network subtyping workflow